Project Report

Date	**29-06-2026**
Project Name	**Automated Network Request Management**
Team ID	
Maximum Marks	

Project Title: Automated Network Request Management in ServiceNow

Project Overview

The Automated Network Request Management system in ServiceNow is developed to simplify and automate the handling of network-related service requests within an organization. These requests typically include VPN access, IP allocation, firewall changes, device connectivity, and other infrastructure-related configurations. In traditional systems, such requests are processed manually, leading to delays, errors, and lack of transparency.

This project utilizes the ServiceNow platform to create a fully automated workflow where users can submit requests through a self-service catalog. Once submitted, the system automatically validates inputs, routes requests for appropriate approvals, and triggers backend processes without manual intervention. It ensures faster processing, consistent decision-making, and improved service delivery.

By integrating key ServiceNow components such as Service Catalog, Catalog Variables, UI Policies, Flow Designer, Approval Management, custom tables, and notification systems, the solution delivers an efficient end-to-end automation framework. It enhances user experience, ensures policy compliance, maintains proper audit records, and provides real-time visibility into request status throughout the lifecycle.

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1. Introduction

Modern organizations rely on stable and secure network infrastructure to support everyday operations. Employees frequently request services such as VPN access, IP configuration, firewall changes, router and switch setup, network relocation, and device onboarding. Handling these requests manually requires significant time and effort, often leading to delays, inconsistencies, and human errors.

As the organization grows, the number of network-related requests also increases, making it challenging to maintain service level agreements (SLAs), ensure proper approval flow, and track request status effectively. Manual processing also reduces visibility and makes it difficult to standardize workflows across teams.

The Automated Network Request Management system in ServiceNow is designed to solve these challenges by providing a centralized and fully automated request handling platform. It allows users to raise requests through a Service Catalog, after which the system automatically validates inputs, routes approvals, generates records, sends notifications, and tracks the complete lifecycle of each request.

This solution delivers multiple benefits, including faster request fulfillment, standardized and automated workflows, reduced dependency on manual effort, improved transparency, better compliance with organizational policies, accurate tracking of all requests, and enhanced user satisfaction.

2. Project Objectives

The main objective of this project is to design and implement an automated network request management system on the ServiceNow platform that reduces manual effort and speeds up the overall request fulfillment process.

This project focuses on creating a structured and user-friendly system where users can submit network-related requests through a Service Catalog, while the platform automatically manages approvals, notifications, record creation, and workflow execution without manual dependency.

The specific objectives of the project include:

- Develop a centralized Service Catalog for handling all network-related service requests.
- Design interactive and dynamic request forms using Catalog Variables.
- Apply UI Policies to show or hide fields based on user input and selection.
- Create custom tables to efficiently store and manage network request data.
- Configure Flow Designer to automate the complete request lifecycle.
- Implement automated approval workflows for managers and network administrators.
- Enable automated email notifications at each stage of request processing.
- Maintain complete audit trails for tracking all network request activities.
- Improve request processing time and ensure better SLA compliance.
- Design a scalable solution that can support additional network services in the future.

3. Key Features

The proposed system is designed with a set of core features that ensure efficient, automated, and user-friendly management of network requests within ServiceNow.

- A simple and intuitive Service Catalog for submitting all network-related requests.
- Smart dynamic forms that adjust field visibility based on user inputs.
- Use of Catalog Variables to capture all necessary request details in a structured manner.
- Implementation of Variable Sets to reuse common user information across multiple requests.
- Automated approval workflows configured using Flow Designer for seamless processing.
- A custom network request table to store and manage all request data efficiently.

- Automatic generation of backend records upon request submission.
- Email notifications triggered during key stages such as approval and completion.
- Real-time tracking of request status for better transparency.
- Complete approval history maintained for audit and compliance purposes.
- A scalable workflow design that supports future enhancements.
- Reduced dependency on manual processing and administrative effort.
- Improved SLA compliance through faster and standardized processing.

4. Project Milestones

Milestone 1: Service Catalog Setup
Objective
To establish a centralized and user-friendly Service Catalog for managing network-related service requests within the ServiceNow platform.
Activities:

- Analyze the different types of network services to be offered.
- Create a dedicated Service Catalog for network requests.
- Configure appropriate Catalog Categories for better organization.
- Develop Service Catalog Items for various network services such as VPN access, firewall changes, and IP allocation.
- Add clear names, descriptions, and user instructions for each catalog item.
- Organize the catalog structure according to ITSM best practices for easy navigation and efficient request management.

Milestone 2: Form Design & Validation

Objective:

To design dynamic and user-friendly request forms that capture accurate network request information.
Activities:

- Create Catalog Variables to collect request details.
- Configure Variable Sets for reusable user information.
- Apply Catalog UI Policies for dynamic field behavior.
- Configure mandatory fields based on request type.
- Implement client-side validation for accurate data entry.
- Enable dynamic show/hide functionality to simplify the form.

Milestone 3: Workflow & Approval Automation

Objective:

To automate the approval process and ensure requests follow the appropriate workflow before fulfillment.
Activities:

- Configure approval rules based on request type.
- Design automated workflows using Flow Designer.

- Route requests to managers or network administrators for approval.
- Send automated email notifications during approval stages.
- Maintain approval history and audit records for compliance.

Milestone 4: System Testing
Objective
To verify that all components of the automated network request system function correctly and meet business requirements.
Activities:

- Perform unit testing of individual components.
- Test Service Catalog forms and validation rules.
- Validate Flow Designer workflows.
- Verify approval routing and notification processes.
- Execute end-to-end request processing scenarios.
- Conduct User Acceptance Testing (UAT) before deployment

Milestone 5: Deployment & Go-Live
Objective
To deploy the completed solution into the production environment and ensure smooth operation for end users.
Activities:

- Review and validate all application configurations.
- Publish Service Catalog items for organizational use.
- Deploy the application using Update Sets.
- Monitor workflow execution and request processing.
- Provide user guidance and training.
- Monitor system performance and resolve post-deployment issues.

5. ServiceNow Developer Setup

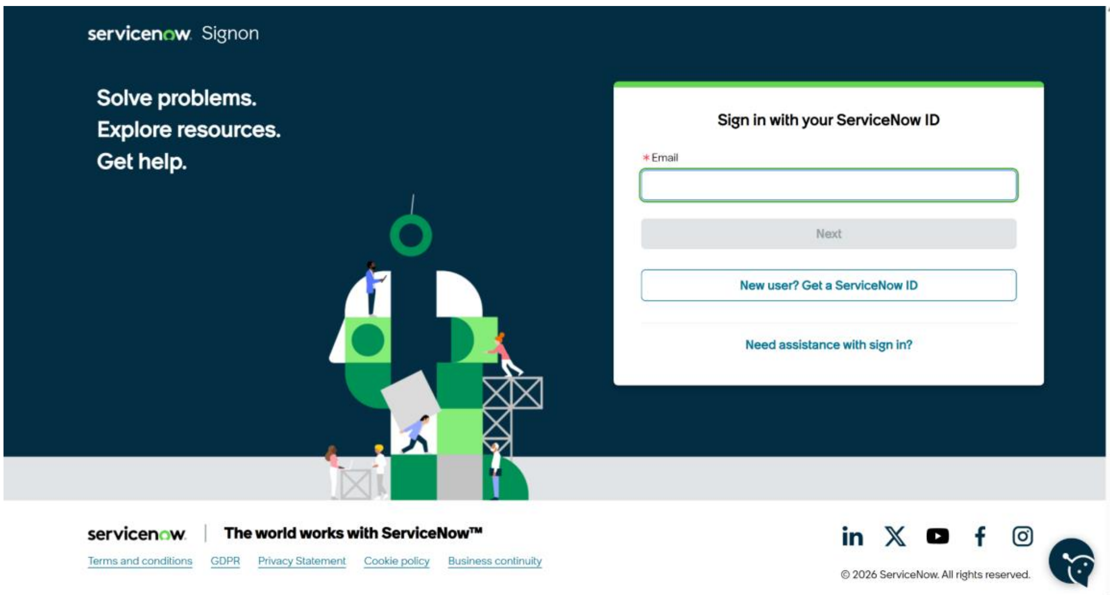
Before developing the application, a ServiceNow Personal Developer Instance (PDI) must be created. The PDI provides a fully functional ServiceNow environment where applications can be developed, tested, and deployed without affecting production systems.

5.1 Creating a ServiceNow Developer Account

To begin development, perform the following steps:

1. Visit the ServiceNow Developer Portal.
2. Click **Sign Up** to create a free developer account.
3. Enter the required information:
    - First Name

    - Last Name
    - Email Address
    - Country
    - Password
4. Verify the email address using the activation link sent to your registered email.
5. Log in to the ServiceNow Developer Portal.
6. Click **Start Building**.
7. Request a **Personal Developer Instance (PDI)**.
8. Wait for the instance to be provisioned.
9. Launch the instance and log in using the generated credentials.



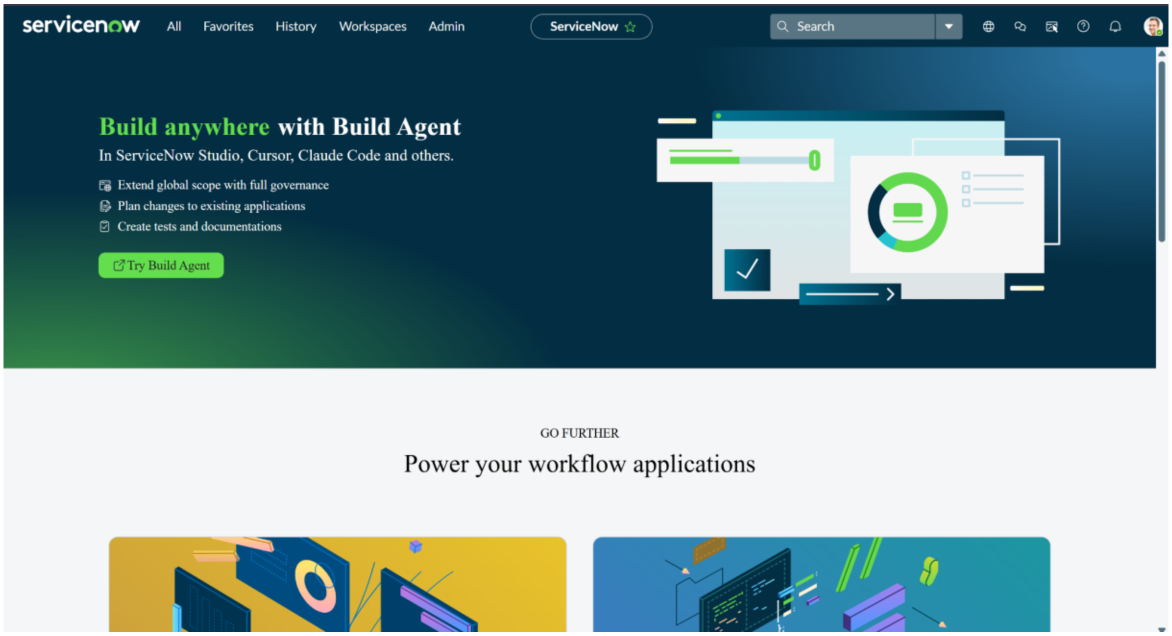
5.2 Accessing Creator Studio

Once the Personal Developer Instance is ready, ServiceNow redirects the developer to Creator Studio.

Creator Studio provides a no-code and low-code environment for building applications. It simplifies application development by allowing developers to create forms, workflows, tables, automation, and integrations through an intuitive graphical interface.

Creator Studio offers features such as:

- Application Creation
- Data Modeling
- Flow Automation
- Form Builder
- Catalog Item Development
- Workflow Configuration



6. Project Implementation in ServiceNow

The implementation of the project consists of multiple modules. Each module contributes to automating the complete network request lifecycle.

The implementation includes:

- Service Catalog Creation
- Catalog Variables Configuration
- Variable Sets
- Catalog UI Policies
- Custom Table Creation
- Field Configuration
- Related Lists
- Flow Designer
- Approval Integration
- Email Notifications
- Record Updates

6.1 Service Catalog Creation

The **Service Catalog** serves as a centralized self-service portal where users can submit network-related requests quickly and efficiently. In this project, a **Network Access Request** catalog item was created to provide users with a single interface for requesting services such as VPN access, firewall changes, IP allocation, and network connectivity.

Steps to Create the Service Catalog Item

1. Open the **Application Navigator** in the ServiceNow developer instance.

2. Search for **Maintain Items** under the Service Catalog module.

3. Navigate to:
   Service Catalog → Catalog Definitions → Maintain Items

4. Click **New** to create a new catalog item.

5. Enter the required details:

   - **Catalog Item Name:** Network Access Request

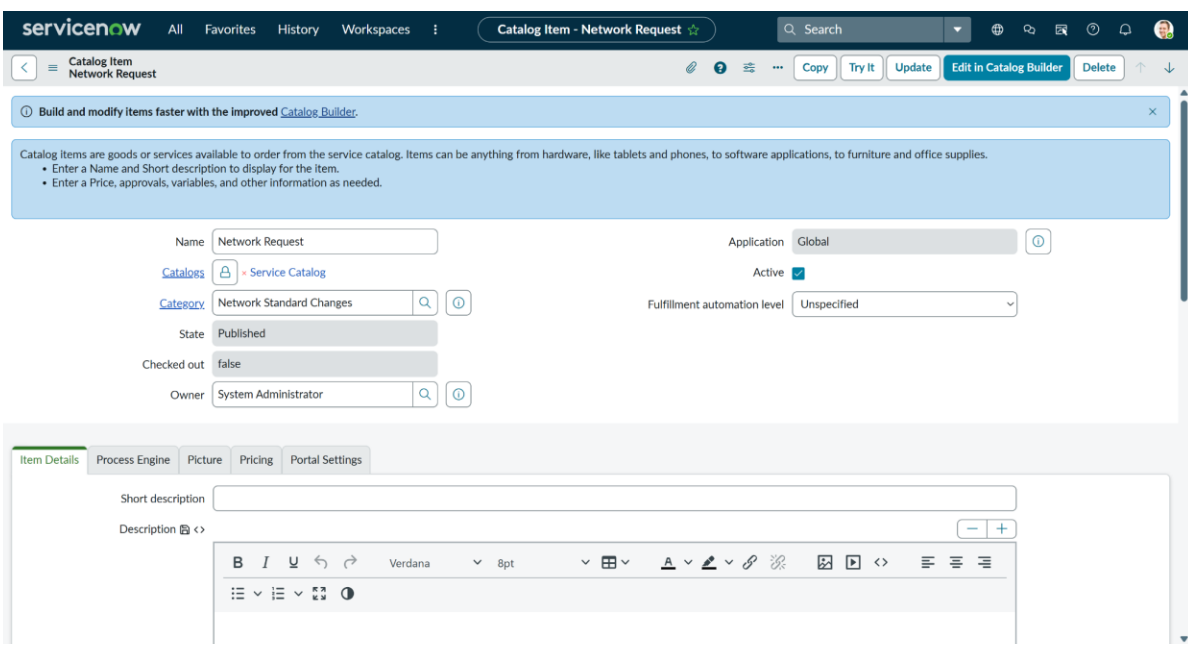
   - **Catalog:** Service Catalog

   - **Category:** Network & Connectivity

   - **Short Description:** Request network access and related infrastructure services through an automated workflow.

6. Configure additional properties such as availability, submission settings, and user access if required.

7. Click **Submit** or **Save** to create the catalog item successfully.



Purpose of the Catalog

The Service Catalog acts as the primary interface through which users can submit network-related service requests in a simple, structured, and standardized manner. It provides a centralized platform that allows employees to request various network services without needing direct interaction with the IT support team.

Users can submit requests for services such as:

- **VPN Access**
- **Firewall Rule Changes**
- **IP Address Allocation**
- **Network Connectivity Support**
- **Device Registration**
- **Network Configuration Changes**

Using the Service Catalog ensures that every request follows a consistent process with complete and accurate information. It eliminates the need for manual emails, phone calls, or paper-based requests, improving efficiency, reducing processing time, and enabling automated approvals, notifications, and request tracking throughout the entire lifecycle.

6.2 Catalog Variables Configuration

Catalog Variables are used to collect information from the requester.

These variables are displayed as fields on the Service Catalog form.

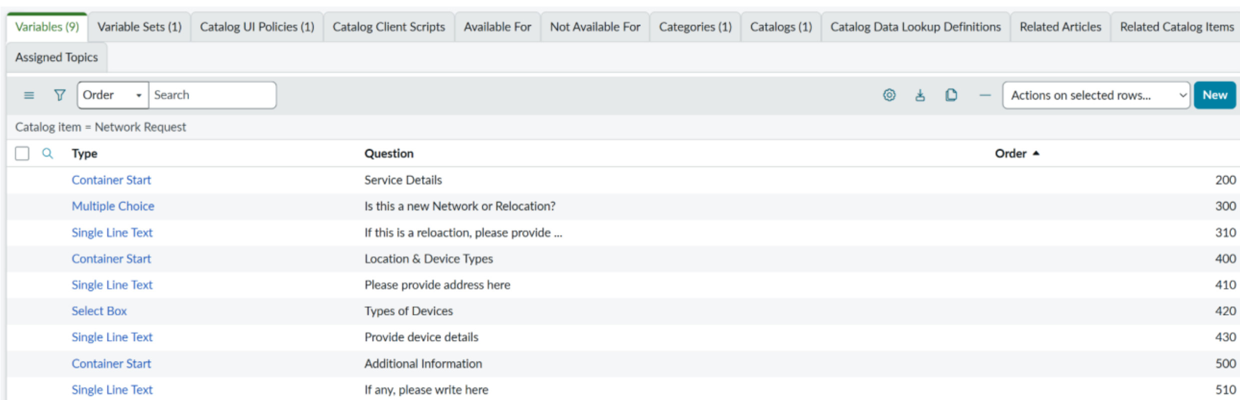
Each variable captures specific information that is later used during workflow automation.

Creating Catalog Variables

Perform the following steps:

1. Open the **Network Request** Catalog Item.
2. Scroll to the **Variables** related list.
3. Click **New**.
4. Configure the following properties:
- Variable Type
- Question
- Name
- Order
- Tooltip
- Help Text
- Mandatory Status
5. Save the variable.

Repeat the process for all required variables.



Variable Configuration

The Service Catalog form uses different types of variables to collect complete and accurate information required for processing network access requests. These variables are organized into logical sections using containers, making the form easy to understand and complete.

Variable Name	Type
Request Details	Container Start
Request Type (VPN Access, Firewall Change, IP Allocation, Network Access)	Multiple Choice
Business Justification	Single Line Text
User & Location Information	Container Start
Department	Select Box
Office Location	Single Line Text
Device Type	Select Box
Device Name / Asset ID	Single Line Text
Network Information	Container Start
IP Address	Single Line Text

Variable Name	Type
Access Duration	Select Box
Additional Comments	Multi Line Text

Purpose of Variable Configuration
The configured variables ensure that users provide all the required information in a structured format. Using containers, multiple-choice fields, select boxes, and text fields improves data accuracy, simplifies form completion, and supports automated validation, approval workflows, and faster request processing within ServiceNow.

Variable Types

1. Reference Variable

Used to reference records from another table.

Example:

Opened On Behalf Of → User Table

This enables automatic retrieval of user information.

2. Choice Variable

Displays predefined options.

Example:

Network Request Type

Options:

- New Connection
- Relocation
- Maintenance
- Configuration Change

3. Single Line Text

Captures short information.

Examples:

- Email

- Username
- Address
- Phone Number

4. Multi-Line Text

Used for detailed descriptions.

Example:

Device Details

Purpose:

Allows users to describe the device or issue in detail.

5. Attachment Variable

Allows users to upload supporting documents.

Examples:

- Identity Proof
- Device Purchase Invoice
- Approval Documents

And many more.

Auto-Populated Variables

Certain variables are automatically populated using **Dot Walking**.

When the user selects **Opened On Behalf Of**, the following fields are filled automatically:

- Email Address
- Username
- Phone Number

This reduces manual data entry and minimizes errors.

Benefits of Catalog Variables

- Standardized data collection
- Improved data accuracy
- Easier automation
- Better reporting

- Reduced manual validation

6.3 Variable Set Configuration

Variable Sets are reusable groups of variables.

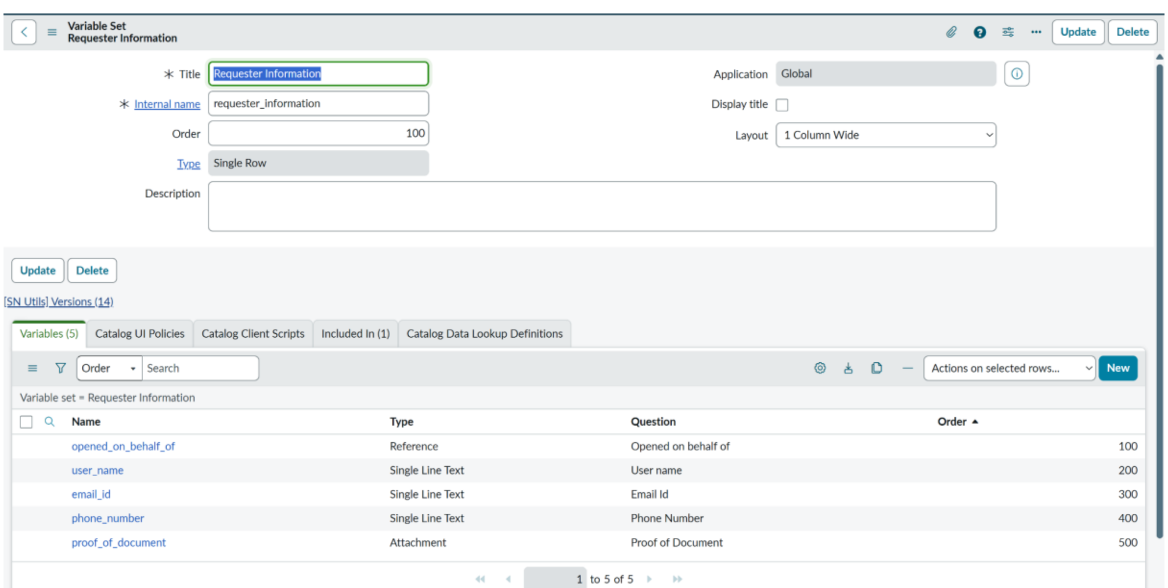
Instead of creating common variables repeatedly for multiple catalog items, Variable Sets allow developers to define them once and reuse them wherever required.

Examples of reusable fields include:

- Opened On Behalf Of
- Email
- Username
- Phone Number
- Document Upload

Creating Variable Sets

1. Navigate to Service Catalog → Variable Sets.
2. Click **New**.
3. Provide:
- Name
- Description
4. Add all required variables.
5. Save the Variable Set.
6. Associate the Variable Set with the Network Request Catalog Item.



6.4 Catalog UI Policy Configuration

Catalog UI Policies improve the user experience by dynamically controlling the visibility, mandatory status, and read-only state of form fields based on user input.

In this project, UI Policies are implemented to display only the relevant fields depending on the type of network request selected by the user. This minimizes unnecessary inputs and ensures that only required information is collected.

Objective

The primary objective of implementing Catalog UI Policies is to:

- Improve form usability.
- Reduce user confusion.
- Display fields dynamically.
- Enforce mandatory fields only when required.
- Prevent invalid or incomplete submissions.

Scenario

If the user selects **Device Type = Others**, an additional field named **"Please Specify Device"** becomes visible.

Similarly, if the user selects **Relocation** as the request type, the **Relocation Address** field is displayed.

Steps to Configure Catalog UI Policy

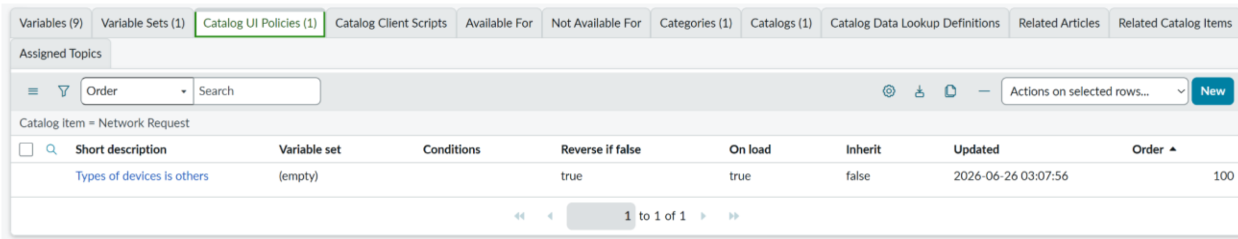
1. Navigate to Service Catalog → Catalog Definitions → Maintain Items.
2. Open the **Network Request** Catalog Item.
3. Scroll down to the **Catalog UI Policies** related list.
4. Click **New**.
5. Enter the following information:
    - **Applies To:** Catalog Item
    - **Catalog Item:** Network Request
    - **Description:** Dynamic Field Visibility
6. Configure the condition.

Example:

Device Type is Others

7. Save the UI Policy.

8. Create a **UI Policy Action**.
9. Select the field **Please Specify Device**.
10. Set **Visible = True**.
11. Update the UI Policy.
12. Test the Catalog Item.



6.5 Creation of Custom Table

A custom table was created to store all submitted network requests.

Instead of relying only on catalog requests, maintaining a dedicated table allows administrators to manage records efficiently, generate reports, and monitor request progress.

Table Details

Table Name: Network Database

Purpose: Store all submitted network requests.

Steps to Create Table

1. Navigate to

System Definition → Tables

2. Click **New**.
3. Enter the following information:
- Label
- Table Name
- Application
4. Enable **Auto Generate Schema**.
5. Click **Submit**.

6.6 Field Creation

After creating the table, fields were added to store request information.

Each field corresponds to one Catalog Variable.

Steps

1. Open the custom table.
2. Navigate to the **Columns** tab.
3. Click **New**.
4. Select the appropriate field type.
5. Enter the Column Label.
6. Provide the Column Name.
7. Save the field.

Repeat the process until all required fields are created.

Fields Created

Field Name	Data Type
Request Number	String
Requested For	Reference
Work Status	Choice
Device Type	Choice
Device Details	String
Customer Address	String
Assignment Group	Reference
Date of Enquiry	Date
Phone Number	String
Email ID	String

Approval Status	Choice

6.7 Request Approval Related List

Approvals are an essential part of network request management.

To maintain approval history, a Related List was configured between the custom table and the Approval table.

Creating Relationship

Navigate to:

System Definition → Relationships

Click **New**.

Configure the following:

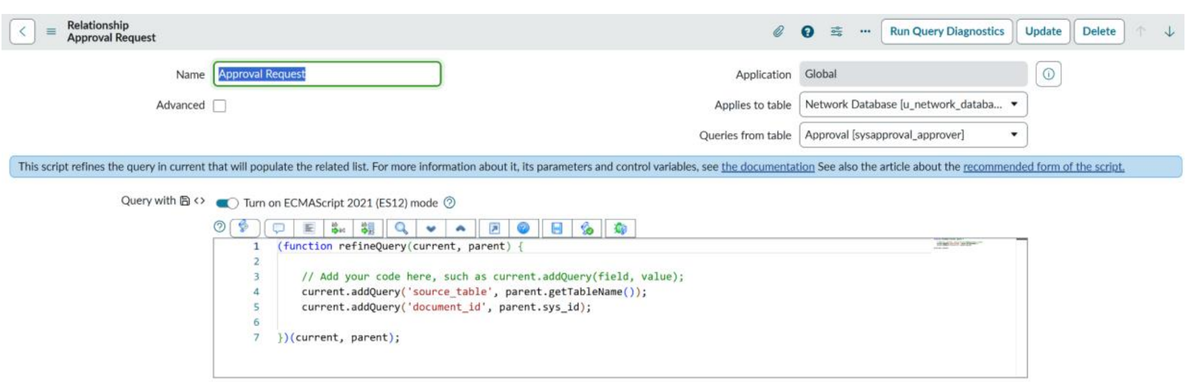
Relationship Name: Approval Request

Applies To: Network Database

Queries From: sysapproval_approver

Active: True

Save the relationship.



Adding Related List

1. Open Form Designer.
2. Select the Network Database table.
3. Add a Related List.
4. Choose **Approval Requests**.
5. Save the form.

Benefits

- Displays approval history.
- Tracks approver decisions.
- Maintains audit records.
- Improves transparency.

6.8 Flow Designer Implementation

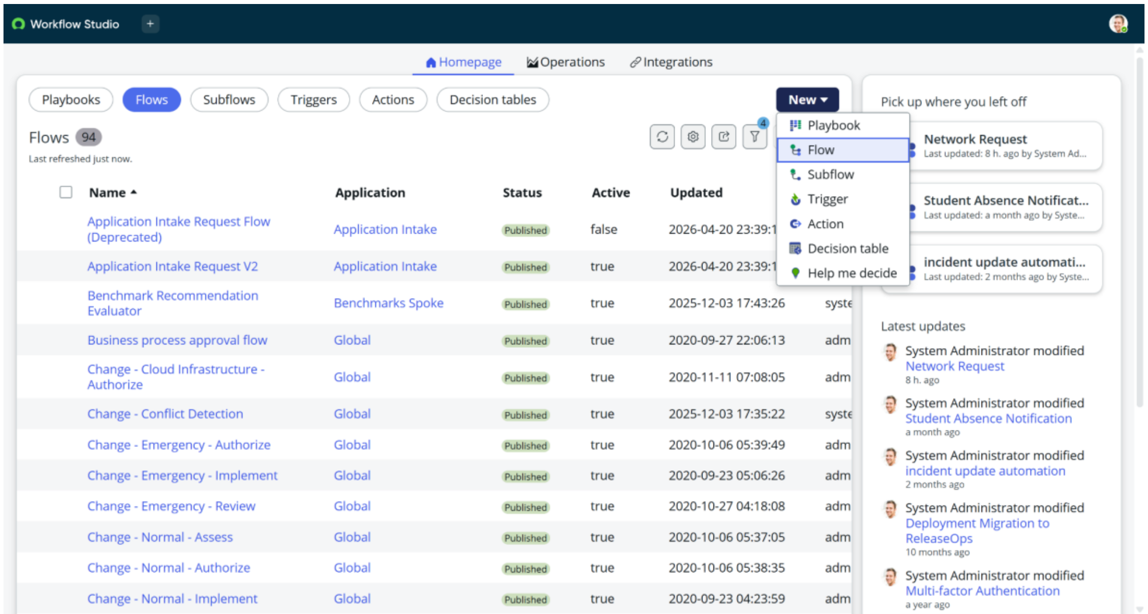
Flow Designer is the core automation engine used in this project. It enables the automatic execution of predefined actions whenever a network request is submitted through the Service Catalog. The flow eliminates manual intervention by creating records, requesting approvals, sending notifications, and updating request statuses based on approval outcomes.

6.8.1 Flow Creation

The first step is to create a new flow that automates the complete network request lifecycle.

Steps to Create the Flow

1. Navigate to **Flow Designer** from the Application Navigator.
2. Click **New**.
3. Select **Flow**.
4. Enter the following details:
    - **Flow Name:** Network Request Management
    - **Description:** Automates the processing of network service requests.
5. Click **Build Flow**.



6.8.2 Configuring the Trigger

The trigger defines when the flow should start.

In this project, the flow is triggered whenever a user submits the **Network Request** catalog item.

Steps

1. Click the **+** icon.
2. Select **Trigger**.
3. Choose **Service Catalog**.
4. Select **Requested Item Created**.
5. Select the **Network Request** catalog item.
6. Save the trigger.

6.8.3 Get Catalog Variables

After the flow starts, all information entered by the user is retrieved using the **Get Catalog Variables** action.

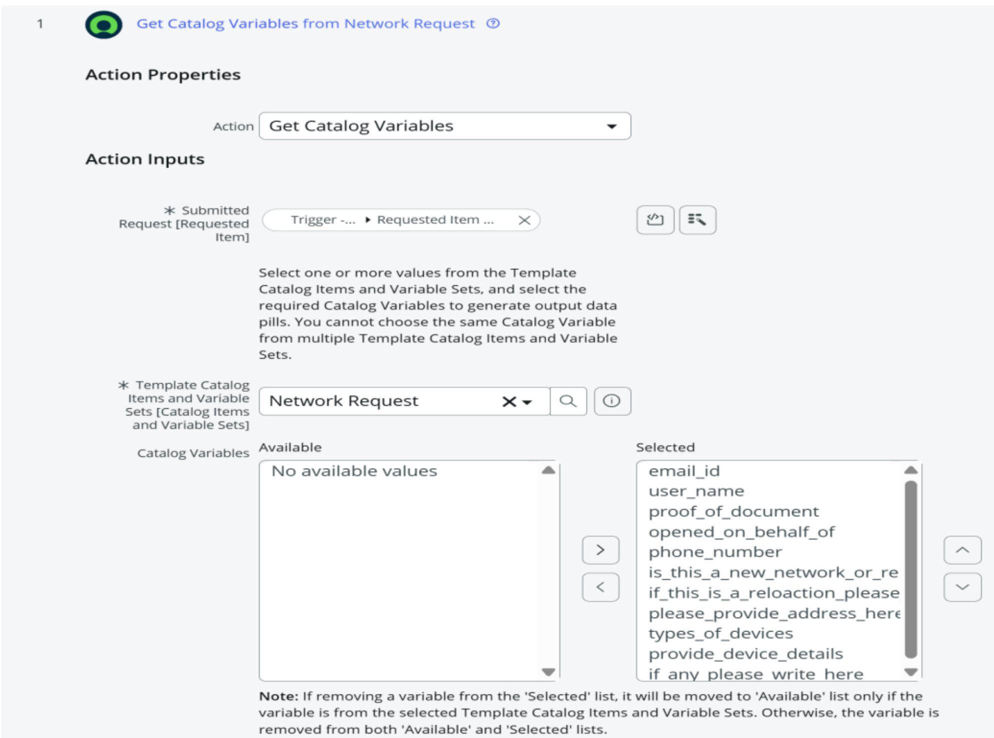
Steps

1. Click **Add Action**.
2. Search for **Get Catalog Variables**.
3. Select the action.
4. Configure the Requested Item as the input.

5. Select all required variables.
6. Save the configuration.

This action retrieves details such as:

- Requested For
- Email ID
- Phone Number
- Device Type
- Device Details
- Address
- Request Type



6.8.4 Create Record

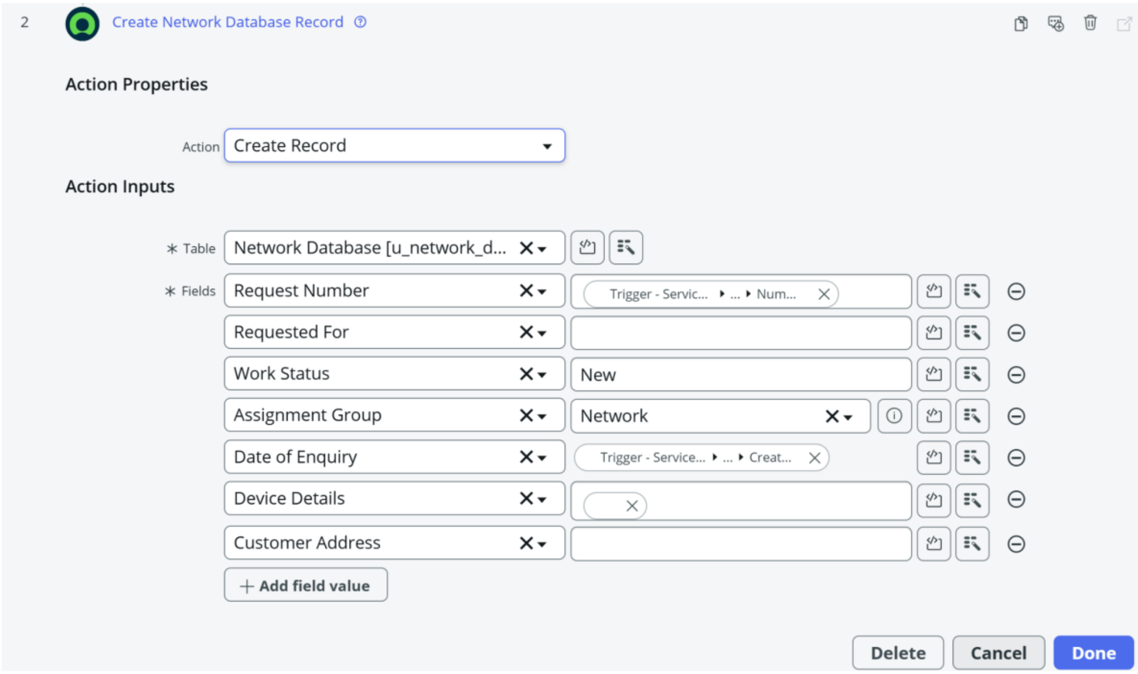
The retrieved information is stored in the custom **Network Database** table.

Steps

1. Add the **Create Record** action.
2. Select the **Network Database** table.
3. Map the catalog variables to the corresponding table fields.

4. Save the action.

The record contains all relevant request information, allowing administrators to monitor and process requests efficiently.



6.8.5 Approval Process

To ensure that only authorized requests are fulfilled, an approval process is integrated into the workflow.

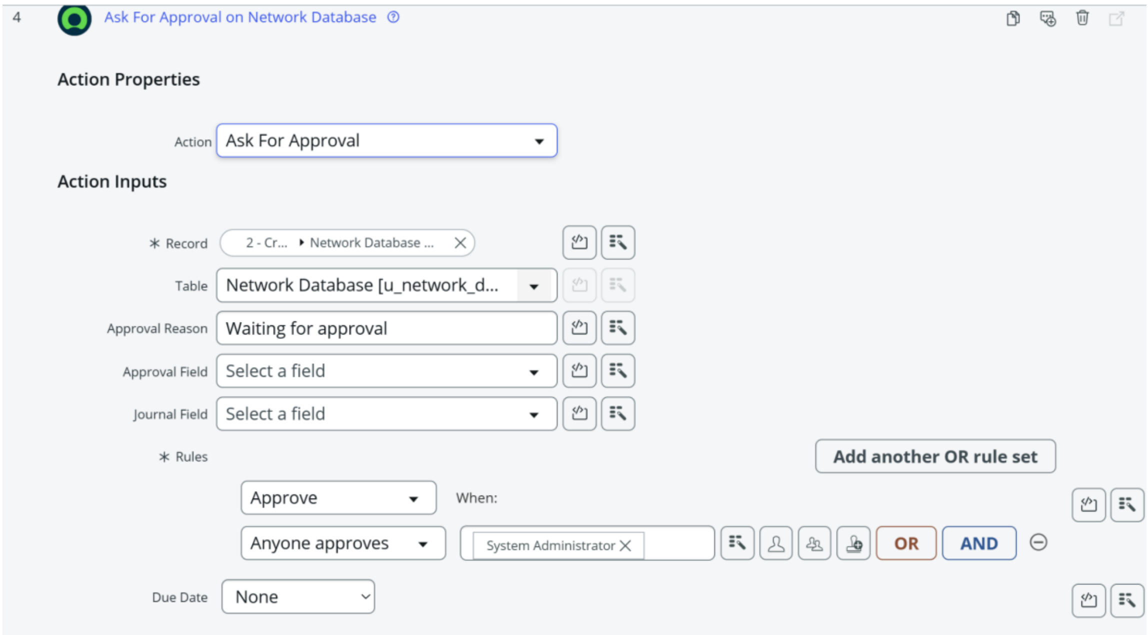
Steps

1. Add the **Ask for Approval** action.
2. Select the newly created Network Database record.
3. Configure the approval rule.
4. Specify the approver (Manager or Network Administrator).
5. Set the approval reason.
6. Save the configuration.

Possible approval outcomes include:

- Approved
- Rejected
- Pending

The approval history is automatically maintained in the system.



6.8.6 Flow Logic

Conditional logic is implemented to determine the next action based on the approval result.

Approved

If the request is approved:

- Send approval email.
- Update the request status to **Approved**.
- Continue with request fulfillment.

Rejected

If the request is rejected:

- Send rejection email.
- Update the request status to **Rejected**.
- Stop further processing.

This ensures that only approved requests proceed through the workflow.

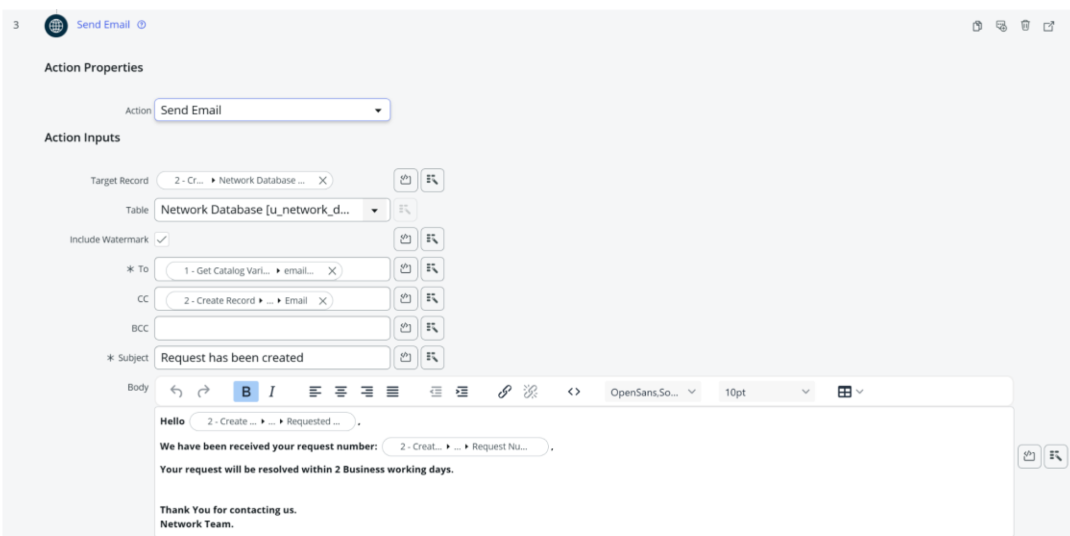
6.8.7 Email Notifications

Automatic email notifications keep users informed about the status of their requests.

The system sends notifications at different stages, including:

- Request Submitted
- Approval Pending
- Request Approved
- Request Rejected
- Request Completed

Each email contains the request details and current status, improving communication and transparency.



6.8.8 Update Record

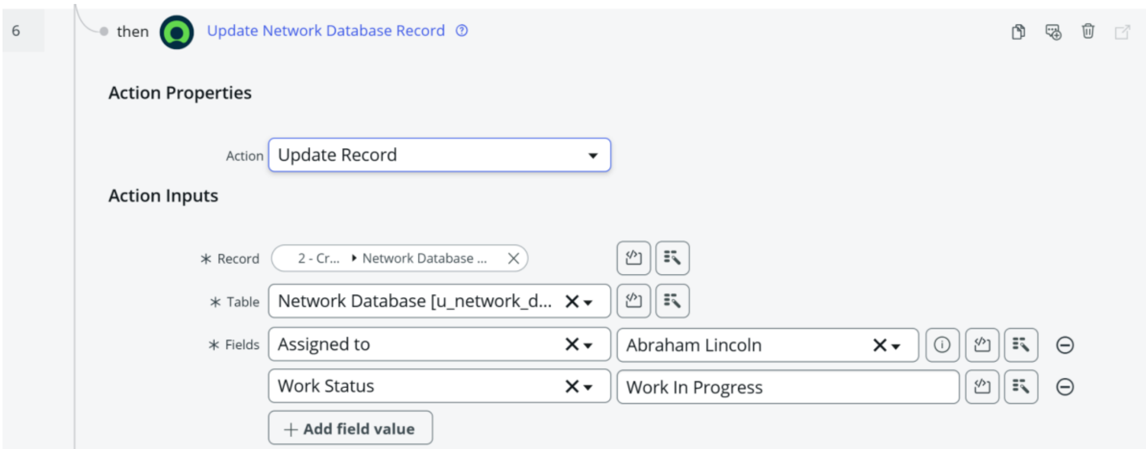
Once the approval process is complete, the corresponding record in the Network Database table is updated automatically.

The system updates information such as:

- Approval Status

- Work Status
- Completion Date
- Assigned Team

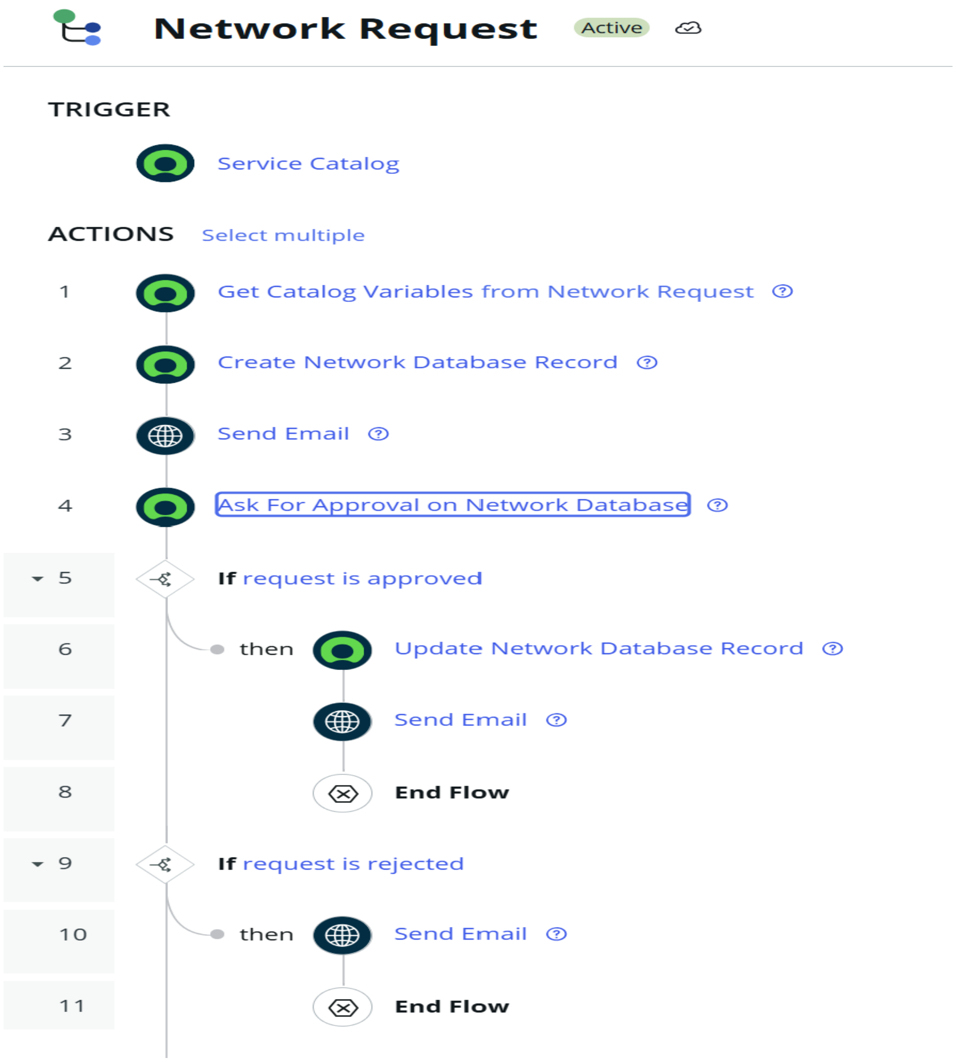
This ensures that the database always reflects the latest status of each request.



6.9 Workflow Summary

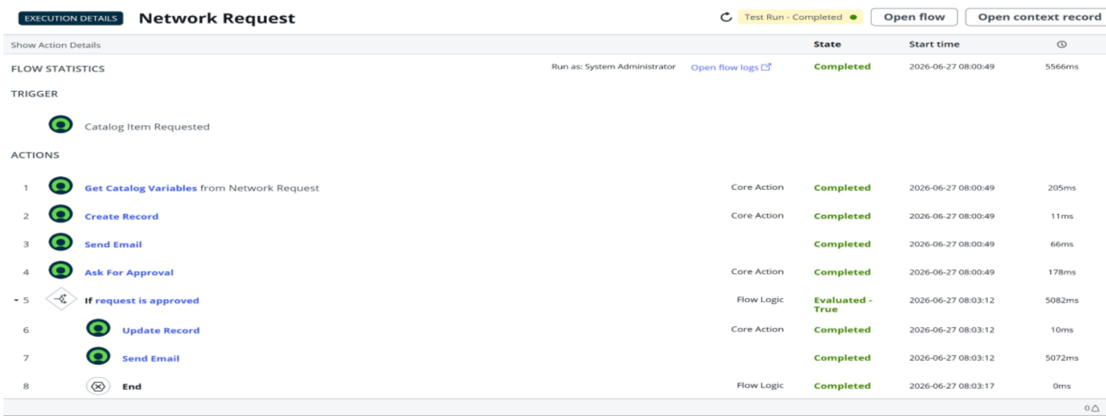
The complete workflow of the project is illustrated below.

1. User submits a Network Request through the Service Catalog.
2. Flow Designer is triggered automatically.
3. Catalog Variables are retrieved.
4. A record is created in the Network Database table.
5. Approval request is sent to the designated approver.
6. The approver reviews the request.
7. Based on the approval decision:
    - If Approved, the request proceeds and the user is notified.
    - If Rejected, the request is closed and the user is informed.
8. The record is updated with the latest request status.

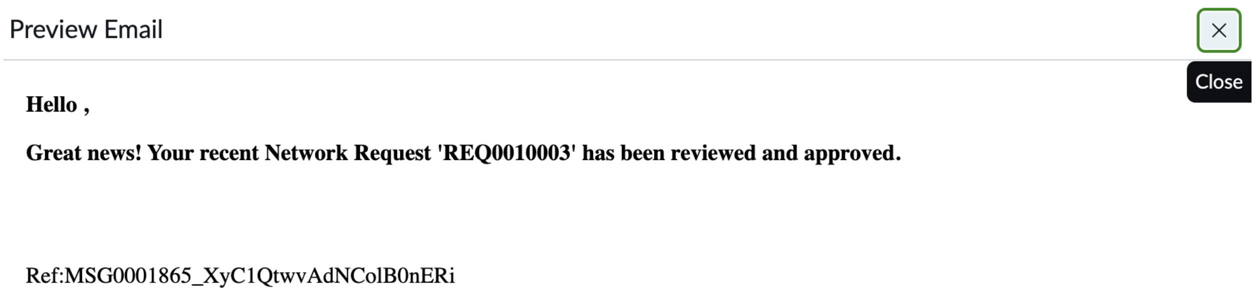


7. Screenshots of Output

1. **Flow Execution Test**



2. **Email Notification**



8. Conclusion

The **Automated Network Request Management** project successfully demonstrates how the ServiceNow platform can automate and simplify the complete lifecycle of network service requests. By utilizing key ServiceNow features such as the **Service Catalog, Catalog Variables, UI Policies, Flow Designer, Approval Workflows, Custom Tables, and Email Notifications**, the system provides a standardized and efficient process for managing network requests from submission through fulfillment.

The automated workflow reduces manual intervention, speeds up request processing, and ensures that approvals are handled consistently according to organizational policies. Real-time notifications and request tracking improve transparency, while centralized record management and audit logs enhance accountability and compliance.

Overall, this solution delivers a reliable, efficient, and scalable network request management system that increases operational productivity, strengthens IT service management, and supports digital transformation within the organization.